Supplementary Table 3. Mixed Effects Linear Models coefficients. ST3.Xa and ST3.Xb show model fixed and random effects, respectively. ST3.Xc shows ANOVA test. ST3.Xd and onward show contrasts computed with *emmeans*. STs are ordered as below.

3.1 – **Accuracy**
3.2 – **Response Times**
3.3 – EEG Amplitudes **Cz – N400**
3.4 – EEG Amplitudes **Pz – N400**
3.5 – EEG Amplitudes **Cz – LPC**
3.6 – EEG Amplitudes **Pz – LPC**

Model 3.1: Accuracy (GLMM binomial)

ST3.1a. Fixed effects — Accuracy

ST3.1a. Fixed effects — Accuracy

Predictor	B	SE	95% CI lower	95% CI upper	z	p
(Intercept)	4.765	0.354	4.071	5.458	13.469	<.001
word_typeL2-Remote	-1.755	0.402	-2.543	-0.966	-4.362	<.001
word_typeL2-Recent	-1.811	0.398	-2.592	-1.030	-4.545	<.001
relatednessRelated	-0.573	0.434	-1.424	0.278	-1.319	0.187
word_typeL2-Remote:relatednessRelated	-0.378	0.532	-1.420	0.664	-0.711	0.477
word_typeL2-Recent:relatednessRelated	-1.136	0.523	-2.161	-0.110	-2.171	<.05

ST3.1b. Random effects — Accuracy

ST3.1b. Random effects — Accuracy

Group	Term	SD
item	sd__(Intercept)	1.080
subject	sd__(Intercept)	0.485

ST3.1c. Type II ANOVA (Wald χ^2) — Accuracy

ST3.1c. Type II ANOVA (Wald χ^2) — Accuracy

Effect	Chi²	gl	p
word_type	89.673	2	<.001
relatedness	39.630	1	<.001
word_type:relatedness	5.781	2	0.056

ST3.1d. Contrasts: word_type (OR) — Accuracy

ST3.1d. Contrasts: word_type (OR) — Accuracy

Contrast	OR	SE	z	p adjusted
L1 / (L2-Remote)	6.986	1.873	7.250	<.001

Contrast	OR	SE	z	p adjusted
L1 / (L2-Recent)	10.791	2.864	8.962	<.001
(L2-Remote) / (L2-Recent)	1.545	0.327	2.052	0.1

ST3.1e. Contrasts: word_type | relatedness (OR) — Accuracy

ST3.1e. Contrasts: word_type | relatedness (OR) — Accuracy

Contrast	Relatedness	OR	SE	z	p adjusted
L1 / (L2-Remote)	Unrelated	5.783	2.327	4.362	<.001
L1 / (L2-Recent)	Unrelated	6.116	2.437	4.545	<.001
(L2-Remote) / (L2-Recent)	Unrelated	1.058	0.335	0.177	0.983
L1 / (L2-Remote)	Related	8.439	2.963	6.075	<.001
L1 / (L2-Recent)	Related	19.041	6.569	8.541	<.001
(L2-Remote) / (L2-Recent)	Related	2.256	0.634	2.895	<.05

ST3.1f. Contrasts: relatedness | word_type (OR) — Accuracy

ST3.1f. Contrasts: relatedness | word_type (OR) — Accuracy

Contrast	Word type	OR	SE	z	p adjusted
Unrelated / Related	L1	1.773	0.770	1.319	0.187
Unrelated / Related	L2-Remote	2.588	0.794	3.097	<.01
Unrelated / Related	L2-Recent	5.521	1.614	5.845	<.001

Model 3.2: Response Times (LMM, log-RT)

ST3.2a. Fixed effects — RT

ST3.2a. Fixed effects — RT

Predictor	B	SE	95% CI lower	95% CI upper	gl	t	p
(Intercept)	0.072	0.043	-0.012	0.157	5,583	1.672	0.094
word_typeL2-Remote	0.136	0.012	0.112	0.161	5,583	10.963	<.001
word_typeL2-Recent	0.126	0.012	0.102	0.149	5,583	10.374	<.001
relatednessRelated	-0.054	0.012	-0.077	-0.031	5,583	-4.654	<.001
word_typeL2-Remote:relatednessRelated	0.007	0.018	-0.027	0.042	5,583	0.404	0.686

Predictor	B	SE	95% CI lower	95% CI upper	gl	t	p
word_typeL2-Recent:relatednessRelated	0.020	0.017	-0.014	0.053	5,583	1.147	0.252

ST3.2b. Random effects — RT

ST3.2b. Random effects — RT

Group	Term	SD
subject	(Intercept)	0.207
item	(Intercept)	0.232

ST3.2c. Type II ANOVA (Wald χ^2) — RT

ST3.2c. Type II ANOVA (Wald χ^2) — RT

Effect	Chi²	df	p
word_type	345.698	2	<.001
relatedness	40.895	1	<.001
word_type:relatedness	1.331	2	0.514

ST3.2d. Contrasts: word_type (ratio) — RT

ST3.2d. Contrasts: word_type (ratio) — RT

Contrast	Ratio	SE	gl	t	p adjusted
L1 / (L2-Remote)	0.870	0.008	5,583	-15.903	<.001
L1 / (L2-Recent)	0.873	0.007	5,583	-15.790	<.001
(L2-Remote) / (L2-Recent)	1.004	0.009	5,583	0.475	0.883

ST3.2e. Contrasts: relatedness (ratio) — RT

ST3.2f. Contrasts: relatedness (ratio) — RT

Contrast	Ratio	SE	gl	t	p adjusted
Unrelated / Related	1.046	0.008	5,583	6.309	<.001

Model 3.3: EEG Amplitudes (glmmTMB) Cz – N400

ST3.3a. Fixed effects — EEG

ST3.3a. Fixed effects — EEG

Predictor	B	SE	95% CI lower	95% CI upper	z	p

Predictor	B	SE	95% CI lower	95% CI upper	z	p
(Intercept)	-0.717	0.239	-1.185	-0.248	-2.999	<.01
word_typeL2-Recent	0.756	0.245	0.275	1.237	3.081	<.01
word_typeL2-Remote	0.231	0.213	-0.187	0.648	1.083	0.279
relatednessRelated	0.228	0.229	-0.220	0.677	0.998	0.318
word_typeL2-Recent:relatednessRelated	-0.348	0.345	-1.024	0.328	-1.008	0.313
word_typeL2-Remote:relatednessRelated	0.287	0.316	-0.332	0.906	0.909	0.364

ST3.3b. Random effects — EEG

ST3.3b. Random effects — EEG

Group	Term	SD
subject	sd__(Intercept)	0.825
item	sd__(Intercept)	0.456
Residual	sd__Observation	

ST3.3c. Type II ANOVA (Wald χ^2) — EEG

ST3.3c. Type II ANOVA (Wald χ^2) — EEG

Effect	Chi²	gl	p
word_type	11.643	2	<.01
relatedness	3.272	1	0.07
word_type:relatedness	3.536	2	0.171

ST3.3d. Contrasts: word_type — EEG

ST3.3d. Contrasts: word_type — EEG

Contrast	Estimate	SE	gl	z	p adjusted
L1 - (L2-Recent)	-0.582	0.173	Inf	-3.371	<.01
L1 - (L2-Remote)	-0.374	0.158	Inf	-2.366	<.05
(L2-Recent) - (L2-Remote)	0.208	0.169	Inf	1.232	0.434

ST3.3e. Contrasts: relatedness | word_type — EEG

ST3.3e. Contrasts: relatedness | word_type — EEG

Contrast	Word type	Estimate	SE	gl	z	p adjusted

Contrast	Word type	Estimate	SE	gl	z	p adjusted
Unrelated - Related	L1	-0.228	0.229	Inf	-0.998	0.318
Unrelated - Related	L2-Recent	0.119	0.258	Inf	0.462	0.644
Unrelated - Related	L2-Remote	-0.515	0.218	Inf	-2.366	<.05

Model 3.4: EEG Amplitudes (glmmTMB) Pz – N400

ST3.4a. Fixed effects — EEG

ST3.4a. Fixed effects — EEG

Predictor	B	SE	95% CI lower	95% CI upper	z	p
(Intercept)	0.910	0.222	0.475	1.346	4.099	<.001
word_typeL2-Recent	1.059	0.237	0.594	1.524	4.465	<.001
word_typeL2-Remote	0.652	0.234	0.193	1.111	2.786	<.01
relatednessRelated	0.048	0.244	-0.430	0.526	0.196	0.844
word_typeL2-Recent:relatednessRelated	-0.490	0.346	-1.168	0.187	-1.418	0.156
word_typeL2-Remote:relatednessRelated	0.311	0.348	-0.371	0.993	0.894	0.371

ST3.4b. Random effects — EEG

ST3.4b. Random effects — EEG

Group	Term	SD
subject	sd__(Intercept)	0.675
item	sd__(Intercept)	0.001
Residual	sd__Observation	

ST3.4c. Type II ANOVA (Wald χ^2) — EEG

ST3.4c. Type II ANOVA (Wald χ^2) — EEG

Effect	Chi²	gl	p
word_type	29.093	2	<.001
relatedness	0.011	1	0.918
word_type:relatedness	5.390	2	0.068

ST3.4d. Contrasts: word_type — EEG

ST3.4d. Contrasts: word_type — EEG

Contrast	Estimate	SE	gl	z	p adjusted
L1 - (L2-Recent)	-0.814	0.173	Inf	-4.703	<.001
L1 - (L2-Remote)	-0.808	0.174	Inf	-4.637	<.001
(L2-Recent) - (L2-Remote)	0.006	0.174	Inf	0.037	0.999

Model 3.5: EEG Amplitudes (glmmTMB) Cz – LPC

ST3.5a. Fixed effects — EEG

ST3.5a. Fixed effects — EEG

Predictor	B	SE	95% CI lower	95% CI upper	z	p
(Intercept)	0.430	0.189	0.060	0.800	2.277	<.05
word_typeL2-Recent	-0.378	0.208	-0.787	0.030	-1.814	0.07
word_typeL2-Remote	-0.321	0.190	-0.693	0.051	-1.689	0.091
relatednessRelated	-0.422	0.202	-0.818	-0.026	-2.088	<.05
word_typeL2-Recent:relatednessRelated	0.463	0.304	-0.133	1.058	1.523	0.128

Predictor	B	SE	95% CI lower	95% CI upper	z	p
word_typeL2-Remote:relatednessRelated	0.381	0.286	-0.180	0.942	1.332	0.183

ST3.5b. Random effects — EEG

ST3.5b. Random effects — EEG

Group	Term	SD
subject	sd__(Intercept)	0.630
item	sd__(Intercept)	0.329
Residual	sd__Observation	

ST3.5c. Type II ANOVA (Wald χ^2) — EEG

ST3.5c. Type II ANOVA (Wald χ^2) — EEG

Effect	Chi^2	gl	p
word_type	1.625	2	0.444
relatedness	1.625	1	0.202
word_type:relatedness	2.808	2	0.246

ST3.5d. Contrasts: word_type — EEG

ST3.5d. Contrasts: word_type — EEG

Contrast	Estimate	SE	gl	z	p adjusted
L1 - (L2-Recent)	0.147	0.152	Inf	0.966	0.598
L1 - (L2-Remote)	0.130	0.143	Inf	0.909	0.635
(L2-Recent) - (L2-Remote)	-0.017	0.152	Inf	-0.111	0.993

Model 3.6: EEG Amplitudes (glmmTMB) Pz – LPC

ST3.6a. Fixed effects — EEG

ST3.6a. Fixed effects — EEG

Predictor	B	SE	95% CI lower	95% CI upper	z	p
(Intercept)	0.641	0.243	0.164	1.119	2.634	<.01
word_typeL2-Recent	-0.235	0.233	-0.691	0.222	-1.008	0.314
word_typeL2-Remote	-0.240	0.227	-0.685	0.205	-1.057	0.291
relatednessRelated	-0.359	0.243	-0.836	0.118	-1.476	0.14
word_typeL2-Recent:relatednessRelated	-0.004	0.347	-0.684	0.676	-0.012	0.99

Predictor	B	SE	95% CI lower	95% CI upper	z	p
word_typeL2-Remote:relatednessRelated	0.230	0.346	-0.449	0.908	0.664	0.507

ST3.6b. Random effects — EEG

ST3.6b. Random effects — EEG

Group	Term	SD
subject	sd__(Intercept)	0.839
item	sd__(Intercept)	0.397
Residual	sd__Observation	

ST3.6c. Type II ANOVA (Wald χ^2) — EEG

ST3.6c. Type II ANOVA (Wald χ^2) — EEG

Effect	Chi^2	gl	p
word_type	1.901	2	0.387
relatedness	4.026	1	<.05
word_type:relatedness	0.593	2	0.744